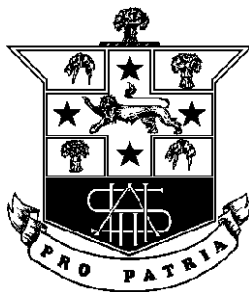


HURLSTONE AGRICULTURAL HIGH SCHOOL



MATHEMATICS – EXTENSION TWO

2008 HSC

ASSESSMENT TASK 1

Examiners ~ G Rawson, J Dillon

GENERAL INSTRUCTIONS

- | | |
|---|---|
| <ul style="list-style-type: none">• Reading Time – 3 minutes.• Working Time – 40 MINUTES.• Attempt all questions.• All necessary working should be shown in every question.• This paper contains two (2) questions. | <ul style="list-style-type: none">• Marks may not be awarded for careless or badly arranged work.• Board approved calculators may be used.• Each question is to be started on a new piece of paper.• This examination paper must NOT be removed from the examination room. |
|---|---|

STUDENT NAME: _____

TEACHER: _____

QUESTION ONE **18 marks** *Start a SEPARATE sheet*

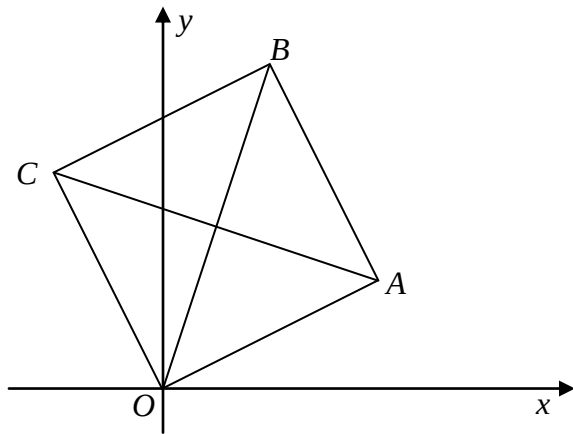
- (a) Evaluate i^{2007} 1
- (b) Simplify $\frac{14+3i}{i(4-5i)}$ 2
- (c) Let z be a non-zero complex number.
If $z^2 = (\bar{z})^2$, show that z is either real or purely imaginary. 2
- (d) Find the complex square roots of $15 + 8i$,
giving your answer in the form $a + ib$, where a, b are real. 3
- (e) Let $z = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$. Find
- (i) $|z|$ 1
- (ii) $\text{Arg}(z)$ 1
- (f) Simplify $\frac{\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)}{\left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12}\right)}$ 3
- (g) Let $z = 1 + \cos \theta + i \sin \theta$ for $0 < \theta < \pi$.
- (i) Show that $z = 2 \cos \frac{\theta}{2} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$ 2
- (ii) Find $|z|$ and $\arg(z)$ in terms of θ . 2
- (h) It is given that $2 + i$ is a root of $P(z) = z^3 + rz^2 + sz + k$.
Under what conditions would $2 - i$ also be a root of $P(z)$? 1

QUESTION TWO 18 marks Start a SEPARATE sheet

- (a) (i) Find the equation of the locus of the point representing the complex number z which satisfies the equation $|z - 2 + i| = 2$. 2
- (ii) Describe this locus geometrically. 1

- (b) Find the locus of the point representing the complex number z which satisfies the equation $|z - 2 + i| = |z + 1 + 3i|$. 2

- (c) The points A , B and C represent the complex numbers z_1 , z_2 and z_3 respectively

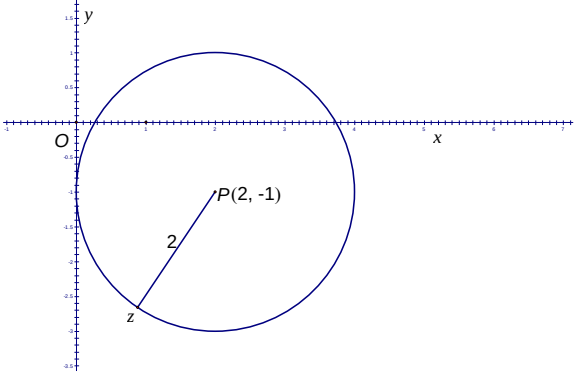
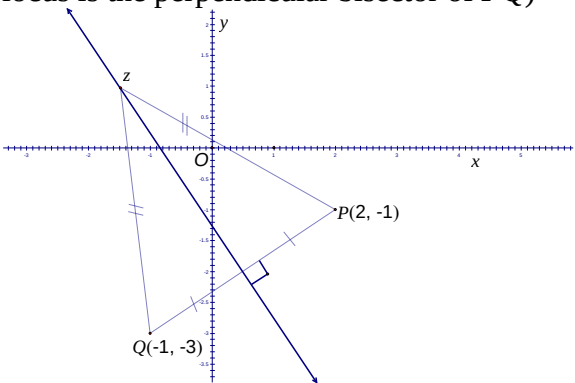


Suppose $OABC$ is a square, and $z_1 = a + ib$,

- (i) Which vectors would represent $z_1 + z_3$ and $z_3 - z_1$? 2
- (ii) Express z_3 and z_2 in terms of a and b . 2
- (ii) Find $\frac{z_3 - z_1}{z_2}$ 2
- (d) Sketch the region in the complex plane where the inequalities $|z - 1 - i| < 2$ and $0 < \arg(z - 1 - i) < \frac{\pi}{4}$ hold simultaneously. 3
- (e) Given $z = x + iy$, sketch the locus of z if $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$.
Show all important features 4

Year 12	Mathematics Extension 2	Ass Task 1 2008 HSC
Question No. 1	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E2	chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings	
E3	uses the relationship between algebraic and geometric representations of complex numbers	
Outcome	Solutions	Marking Guidelines
(a)	$i^{2007} = (i^3)(i^{2004}) = (-i)(1) = -i$	<u>1 mark</u> : correct answer
(b)	$\frac{14+3i}{i(4-5i)} = \frac{14+3i}{5+4i} \times \frac{5-4i}{5-4i}$ $= \frac{70-56i+15i+12}{25+16}$ $= \frac{82-41i}{41}$ $= 2-i$	<u>2 marks</u> : <u>1 mark</u> :
(c)	Let $z = x + iy$ Then $z^2 = \bar{z}^2$ $(x+iy)^2 = (x-iy)^2$ $x^2 - y^2 + 2xyi = x^2 - y^2 - 2xyi$ $xy = -xy$ $xy = 0$ $\Rightarrow x = 0 \text{ or } y = 0$ $\therefore z \text{ is real or imaginary}$	<u>2 marks</u> : <u>1 mark</u> :
(d)	Let $x + iy = \sqrt{15+8i}$ $(x+iy)^2 = 15+8i$ $x^2 - y^2 + 2xyi = 15+8i$ $x^2 - y^2 = 15 \quad \dots(i)$ $2xy = 8 \quad \dots(ii)$ from (ii) $y = \frac{4}{x} \quad \dots(iii)$ sub (iii) $\rightarrow$ (i), we have $x^2 - \left(\frac{4}{x}\right)^2 = 15$ $x^4 - 15x^2 - 16 = 0$ $(x^2+1)(x^2-16) = 0$ $x = \pm 4 \text{ or } \pm i \text{ but } x, y \in \mathbb{R}$ $\therefore x = 4, y = 1$ or $x = -4, y = -1$ ie $\sqrt{15+8i} = 4+i, -4-i$ $= \pm(4+i)$	<u>3 marks</u> : <u>2 marks</u> : <u>1 mark</u> :

(e)(i)	$ z = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(-\frac{\sqrt{2}}{2}\right)^2}$ $= \sqrt{\frac{1}{2} + \frac{1}{2}}$ $= 1$	<u>1 mark</u> : correct solution
(e)(ii)	$\text{Arg}(z) = \tan^{-1}\left(\frac{-\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}}\right)$ $= \tan^{-1}(-1)$ $= -\frac{\pi}{4}$	<u>1 mark</u> : correct solution
(f)	$\frac{\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)}{\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)}$ $= \cos\left(\frac{\pi}{3} + \frac{\pi}{4} - \frac{\pi}{12}\right) + i\sin\left(\frac{\pi}{3} + \frac{\pi}{4} - \frac{\pi}{12}\right)$ $= \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$ $= i$	<u>3 marks</u> : <u>2 marks</u> : <u>1 mark</u> :
(g)(i)	$z = 1 + \cos\theta + i\sin\theta$ $= 1 + \left(2\cos^2\frac{\theta}{2} - 1\right) + i\left(2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)$ $= 2\cos\frac{\theta}{2}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)$	<u>2 marks</u> : <u>1 mark</u> :
(g)(ii)	$ z = 2\cos\frac{\theta}{2} \quad \text{and} \quad \arg z = \frac{\theta}{2}$	<u>2 marks</u> : <u>1 mark</u> :
(h)	Complex roots occur in conjugate pairs when coefficients of $P(z)$ are real. So if $2 + i$ is a root, $2 - i$ will also be a root <u>when r, s and k are all real.</u>	<u>1 mark</u> :

Year 12	Mathematics Extension 2	Ass Task 1 2008 HSC
Question No. 2	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E2	chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings	
E3	uses the relationship between algebraic and geometric representations of complex numbers	
Outcome	Solutions	Marking Guidelines
(a)(i)	Let $z = x + iy$ Then $z - 2 + i = 2$ $ (x + iy) - 2 + i = 2$ $ (x - 2) + i(y + 1) = 2$ $\sqrt{(x - 2)^2 + (y + 1)^2} = 2$ $(x - 2)^2 + (y + 1)^2 = 4$ 	<u>2 marks</u> : <u>1 mark</u> :
	The locus is a circle [centre (2, -1), radius 2]	<u>1 mark</u> :
	(b) Let $z = x + iy$ Then $z - 2 + i = z + 1 + 3i$ $ x + iy - 2 + i = x + iy + 1 + 3i $ $ (x - 2) + i(y + 1) = (x + 1) + i(y + 3) $ $(x - 2)^2 + (y + 1)^2 = (x + 1)^2 + (y + 3)^2$ $x^2 - 4x + 4 + y^2 + 2y + 1 = x^2 + 2x + 1 + y^2 + 6y + 9$ $6x + 4y + 5 = 0$ (Again, this can be done graphically by noting the locus is the perpendicular bisector of PQ) 	<u>2 marks</u> : <u>1 mark</u> :

(c)(i)

$z_1 + z_3$ is represented by $\overrightarrow{OB}$

$z_3 - z_1$ is represented by $\overrightarrow{AC}$

2 marks :

1 mark :

(c)(ii)

$\overrightarrow{OC}$ is $\overrightarrow{OA}$ rotated anticlockwise through 90°

$$\text{so } z_3 = i.z_1$$

$$= i(a + ib)$$

$$= -b + ia$$

$$\text{Also, } z_2 = z_3 + z_1$$

$$= (-b + ai) + (a + bi)$$

$$= (a - b) + (a + b)i$$

(c)(iii)

$$\frac{z_3 - z_1}{z_2} = \frac{(-b + ai) - (a + bi)}{(a - b) + (a + b)i}$$

$$= \frac{(-a - b) + (a - b)i}{(a - b) + (a + b)i}$$

$$= \frac{[-(a + b) + (a - b)i]}{[(a - b) + (a + b)i]} \times \frac{[(a - b) - (a + b)i]}{[(a - b) - (a + b)i]}$$

$$= \frac{[-(a^2 - b^2) + (a^2 - b^2)] + [(a - b)^2 + (a + b)^2]i}{(a - b)^2 + (a + b)^2}$$

$$= i$$

2 marks :

1 mark :

Alternatively... (better method)

$OB = AC$ and $OB \perp AC$ (diagonals of square)

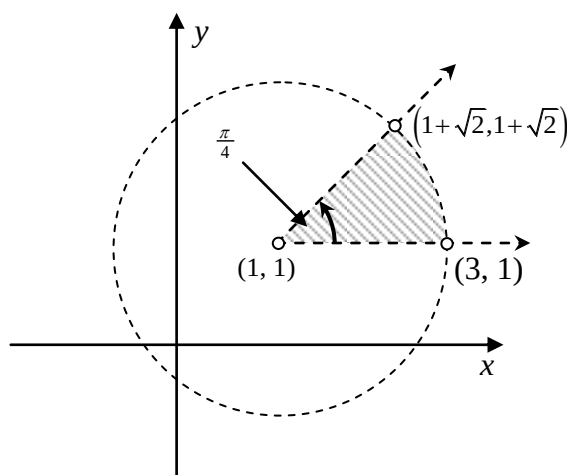
so, $\overrightarrow{AC}$ is obtained by rotating $\overrightarrow{OB}$ anticlockwise through 90° , and then translating (no change in length).

$$\text{ie, } \overrightarrow{AC} = i.\overrightarrow{OB}$$

$$z_3 - z_1 = i.z_2$$

$$\therefore \frac{z_3 - z_1}{z_2} = i$$

(d)



3 marks :

2 marks :

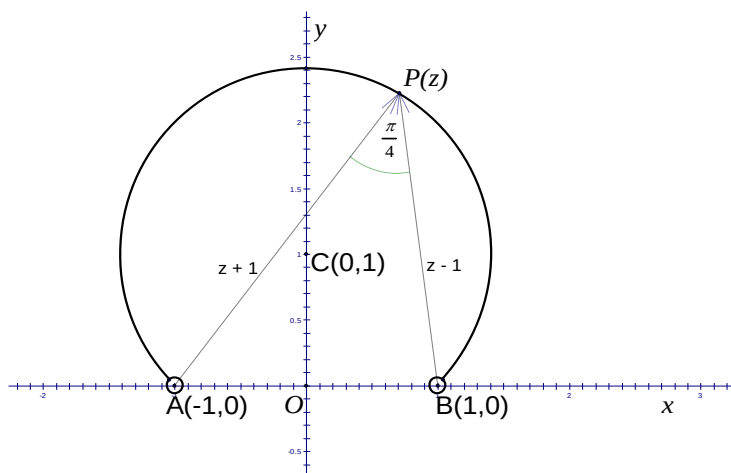
1 mark :

(e)

$$\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$$

$$\arg(z-1) - \arg(z+1) = \frac{\pi}{4}$$

$$\arg(z+1) = \arg(z-1) - \frac{\pi}{4}$$



Algebraically:

$$\frac{z-1}{z+1} = \frac{x+iy-1}{x+iy+1}$$

$$= \frac{[(x-1)+iy]}{[(x+1)+iy]} \cdot \frac{[(x+1)-iy]}{[(x+1)-iy]}$$

$$= \frac{[(x^2-1^2)+y^2] + [y(x+1)-y(x-1)]i}{(x+1)^2+y^2}$$

$$= \frac{[x^2+y^2-1]}{(x+1)^2+y^2} - \frac{2y}{(x+1)^2+y^2}i$$

$$\tan\left[\arg\left(\frac{z-i}{z+i}\right)\right] = \frac{-\frac{2y}{(x+1)^2+y^2}}{\frac{x^2+y^2-1}{(x+1)^2+y^2}}$$

$$= -\frac{2y}{x^2+y^2-1}$$

$$\text{as } \tan\frac{\pi}{4} = 1$$

$$-\frac{2y}{x^2+y^2-1} = 1$$

so the relation is $x^2 + y^2 + 2y - 1 = 0$

Note: this method alone will not show that the locus is above the x-axis only.

4 marks :

3 marks :

2 marks :

1 mark :